

Endpoint Compliance:

- o Endpoint device is an Internet-capable computer hardware device on network.
- o Endpoint refer to desktop computers, laptops, smart phones, tablets, printers etc.
- o Endpoint device is LAN-connected hardware device that communicates across network.
- o **Compliance** is state of being in accordance with established guidelines & specifications.
- o If an endpoint device has been founded **Compliance** with the security policy.
- o Endpoint device will be able to access a larger amount of network resources.
- o If the endpoint is **compliant**, it might be able to access all network resources.
- o It will access network resources including file shares, web servers, email & so on.
- o If an endpoint device has given a posture assessment of **Noncompliant**, state.
- o Endpoint granted minimal level of network access to remediate its noncompliance.
- o This can allow the endpoint to access a webpage to download the latest antivirus.
- o This can allow the endpoint device to access antimalware software and definitions.
- o Alternatively, access to download the appropriate corporate software packages.
- o If an endpoint device has been founded **Unknown** from a posture standpoint.
- o It means that Network Admission Control agent is likely, not installed on endpoint.
- o Alternatively, endpoint was unable to communicate posture assessment back to ISE.
- o The compliance status of an endpoint can be Unknown, Noncompliant & Compliant.
- o **Unknown**: No data was collected from endpoint in order to determine posture state.
- o **Noncompliant**: Posture assessment was performed, & one or more requirements failed.
- o **Compliant**: The endpoint device is compliant with all mandatory requirements.

Client Provisioning:

- o Client Provisioning, feature downloads client provisioning resources.
- o Client Provisioning feature configure Agent profiles at the User end.
- o Configure how each endpoint type will be provisioned to join the network.
- o Client Provisioning sometimes referred to as onboarding the endpoint device.
- o When the endpoint attempts to onboard onto the network to access resources.
- o The appropriate security credentials must be downloaded onto the endpoint.
- o If Client provisioning option is disabled, user will show a warning message.
- o Client Provisioning enabled network session when client connects to the network.
- o When Client attempts to connect the network to access the network resources.
- o Client Provisioning resources are downloaded to the client device.
- o Client Provisioning is processing whereby all credentials and configurations are deployed.
- o Client provisioning, allowing endpoint automatically, join the network in the future.
- o Client provisioning, portion of posture service installs the NAC agent onto the endpoint.
- o NAC Agent is responsible for the checking the Endpoint device for the compliance.
- o This NAC agent is automatically, downloaded when user logs in through portal.

Posture Services:

- o Posture Service make sure configured Network Access Device with Dot1x Authentication.
- o Either Authenticate endpoint device using MAB configured continue to access.
- o To enable Posture services, install license on Cisco Identity Service Engine (ISE).
- o Posture services comprised three main Client provisioning, Posture & Authorization policy.
- o Network Admission Control (NAC) Agents are installed on the client's machines.
- o NAC interact with the posture service to enforce security policies on all the endpoints.
- o These policies apply on all endpoints that attempt to gain access to protected network.
- o Network Admission Control Agents assist in evaluating clients against the posture policies.
- o NAC agents enforcing the security policies on clients to meet the compliance.

Posture Conditions:

- o Posture conditions are used to check specific attributes on the client system.
- o There are several categories for assessment options.
- o These conditions can be used alone or in combination with other conditions:

1. File Conditions:

File Conditions checks existence of a file, the date of a file & versions of file on the client.

2. Registry Conditions:

Registry condition checks for existence of registry key or the value of the registry key on client.

3. Application Conditions:

A condition that checks if an application (process) is running or not running on the client.

4. Service Conditions:

A condition that checks if a service is running or not running on the client.

5. Dictionary Conditions:

A condition that checks a dictionary attribute with a value.

1. Compound Conditions:

Contains one or more simple, or compound conditions type File, Registry, Application or Service

2. Antivirus Compound Conditions:

Contains one or more AV conditions, or AV compound conditions

3. Antispyware Compound Conditions:

Contains one or more AS conditions, or AS compound conditions

4. Dictionary Compound Conditions:

Contains one or more dictionary simple conditions or dictionary compound conditions